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SUMMARY

Robotics-focused NYU M.S. candidate with hands-on experience building real-world autonomous systems across perception, planning, control, and embedded robotics (ROS 2, computer vision, and microcontrollers).

EXPERIENCE

Research Assistant - NYU Mechatronics, Controls and Robotics Lab <i>VLM-LLM Integrated Multi-Robot System for Autonomous Task Execution</i>	June 2025 - Present
- Architected a zero-code LLM-VLM pipeline using Gemini 2.5 Pro for high-level reasoning, deconstructing natural language into executable tool-calls for complex sorting tasks. - Engineered a full-autonomy stack: multi-camera CV perception (<1cm accuracy) → RRT* motion planning (online replanning) → Pure Pursuit+PID for precise navigation and gripper docking. - Validated system performance via real-robot experiments, achieving zero collisions during dynamic multi-robot coordination and autonomous obstacle avoidance iterations. - Orchestrated a ZeroMQ/CBOR RPC layer with a multi-comm bridge (Wi-Fi/BLE/ESP-NOW), ensuring low-latency command execution between the high-level reasoning agent and hardware.	
Course Assistant – Automatic Controls Laboratory <i>New York University, Tandon School of Engineering</i>	Sep 2025 – Present
- Spearheaded Automatic Controls labs using Quanser and MATLAB-Simulink, implementing real-time PID and LQR control for high-order dynamical systems to achieve precise set-point tracking. - Redesigned curriculum via Arduino-based embedded labs, utilizing system identification and digital filter design to bridge the gap between continuous-time theory and discrete-time hardware.	
AI Systems Engineer <i>Lawyantra, Mumbai, India</i>	Dec 2023 – May 2024
- Constructed a high-performance legal search engine using RAG architecture and Pinecone hybrid search, achieving 99% relevance scores for document retrieval across a database of 20,000+ international court records. - Pioneered a multi-tiered summarization pipeline leveraging Llama 2, Mistral, and GPT-3.5 Turbo, successfully condensing 1,000-token legal PDFs into 50-token summaries while maintaining critical legal essence. - Optimized retrieval precision by integrating TW-Bert and BM25 algorithms, utilizing vector embeddings and term-frequency scoring to rank most-relevant case law with near real-time response times.	
Mechanical Design & Validation Engineer (SAE BAJA) <i>MSPTME Racing Team</i>	Sep 2021 – May 2022
- Designed an ATV hydraulic braking system using Fusion360, optimizing component geometry to achieve a 15% reduction in subsystem weight. - Analyzed high-stress loading conditions via FEA (Ansys), achieving a 96% correlation between simulation data and field trial results to enhance subsystem reliability. - Integrated the braking subsystem as a vehicle-level assembly, validating performance across 40+ dynamic test runs to secure a 10th place rank in SAE BAJA Phase 2.	

PROJECTS

Interactive Motion-Controlled Rehabilitation Game	April 2025– May 2025
- Developed a multimodal motion-tracking stack using OpenCV HSV filtering and dual MPU6050 IMUs, achieving sub-100ms end-to-end latency. - Formulated a sensor fusion pipeline on Raspberry Pi 400 that processes 200Hz acceleration data to compute numerical jerk derivatives for 95% detection reliability. - Streamlined a wireless communication bridge using the ESP-NOW protocol, reducing transmission overhead by 40% compared to standard WiFi.	
Indoor Mapping Robot with Real-Time ROS Visualization	Jan 2025 - Feb 2025
- Produced a differential drive mapping robot using ROS2, integrating a custom 360-degree ultrasonic LiDAR to generate point cloud visualizations in RViz at 10Hz. - Executed a multi-sensor fusion pipeline using a Complementary Filter (75% IMU, 25% Encoder), reducing orientation drift by 15% for stable odometry. - Deployed a distributed communication architecture utilizing Micro-ROS on an ESP32, achieving sub-50ms latency in bidirectional data transfer.	

EDUCATION

New York University, New York, NY <i>Master of Science, Mechatronics and Robotics</i>	May 2026 GPA: 3.61
NMIMS University, Mumbai, India <i>Bachelor of Technology, Mechatronics</i>	May 2024 GPA: 3.69

SKILLS

Languages: Python, C/C++, MATLAB Robotics: ROS 2, Localization, SLAM, Path Planning, Kinematics & Dynamics, Sensor Fusion (IMU/Encoders/Vision) AI & Perception: OpenCV, ArUco Detection, PyTorch, Vision-Language Models, LLM Integration, RAG, Transformers (BERT) Controls: PID, LQR, System Identification, State Estimation, Model-Based Design, Real-Time Feedback Control Embedded Systems: ESP32, Arduino, Raspberry Pi, Jetson; I2C/UART, SPI, Wi-Fi, BLE, ESP-NOW; PLC (IndraWorks), SCADA/HMI Tools & CAD/CAE: Git/GitHub, Linux (Ubuntu), SolidWorks, Fusion 360, Ansys, 3D Printing, Laser Cutting	
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